MongoDB Architecture and Characteristics

Aditi · Follow
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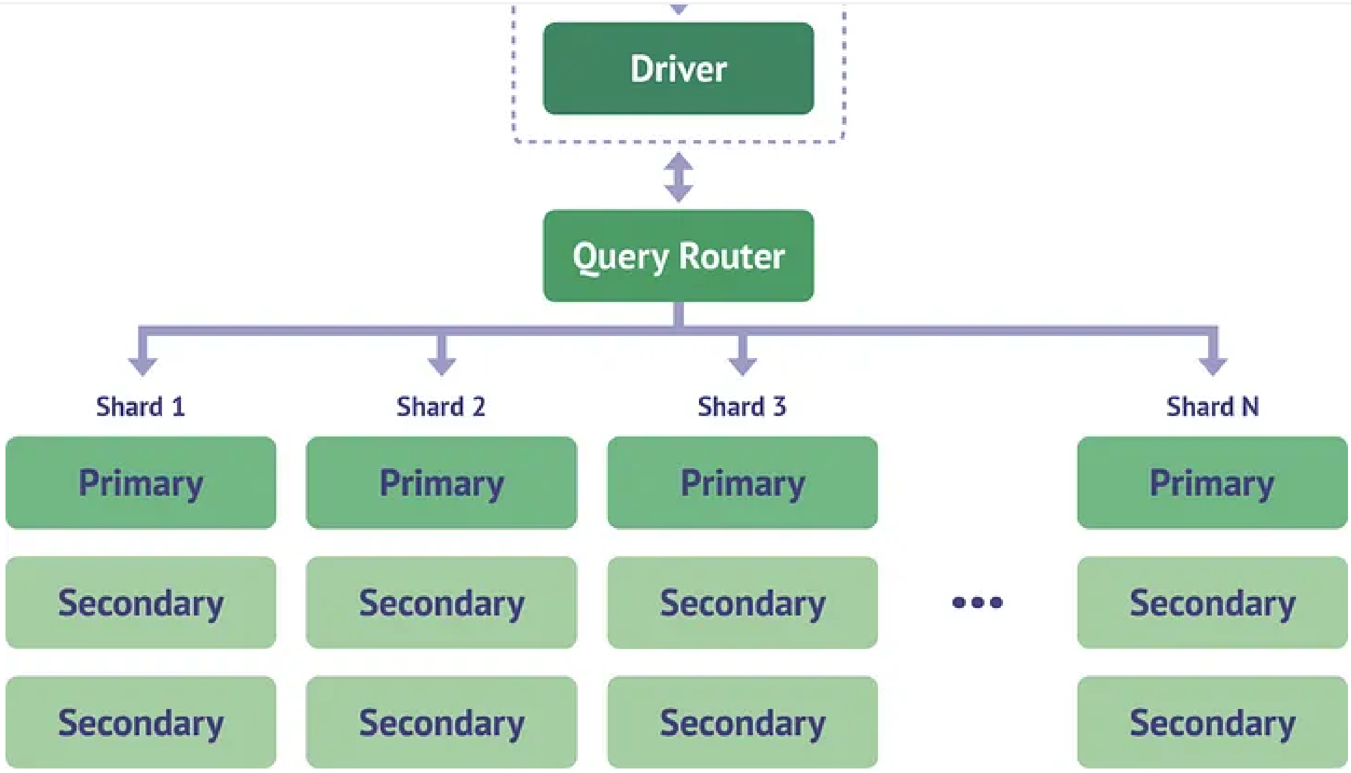
MongoDB is an open-source, cross-platform, document-oriented NoSQL database. It is designed to store and process large amounts of semi-structured or unstructured data, such as JSON documents. It is written in C++.

Important characteristics of MongoDB

1. **Document-Oriented:** MongoDB uses a document data model, which means that data is stored in flexible, JSON-like documents instead of in a rigid, table-based structure. This allows for greater flexibility and ease of use, as the structure of the data can change over time without affecting the overall system.

2. **High Scalability:** MongoDB uses sharding to horizontally partition data across multiple servers, making it easy to scale out as the amount of data grows.
3. **High Availability:** MongoDB provides built-in replication and automatic failover capabilities, ensuring that data remains available even in the event of a node failure.
4. **Optimized for heavy reads**: MongoDB uses a leader-follower protocol, making it possible to use replicas for heavy reading.
5. **Strong Consistency**: MongoDB provides strong consistency through its replica sets, ensuring that all members of a replica set have the same data.
6. **Atomicity at the document level**: MongoDB ensures atomicity in concurrent write operations at the document level. This means that write operations on a single document are guaranteed to be atomic and isolated from each other, even when performed concurrently.
7. **Rich Query Language:** MongoDB supports a rich and powerful query language that allows you to retrieve and manipulate data with ease.
8. **Flexible Data Modeling:** MongoDB allows for flexible and dynamic data modeling, allowing for changes to the schema without affecting the overall system.
9. **Indexing**: MongoDB supports a wide range of indexing options, including secondary indexes, text search, and geospatial indexing, making it easy to retrieve data quickly.
10. **Built-in Aggregation Framework**: MongoDB provides a built-in aggregation framework for data analysis and reporting.
11. **Community-Driven:** MongoDB has a large and active community of users, developers, and contributors, making it a well-supported and widely used database technology.

MongoDB Architecture



The MongoDB architecture consists of the following components:

1. **Collection:** A collection is a group of MongoDB Documents, which is equivalent to an RDBMS table.
2. **Document:** A document is a set of key-value pairs, where a value can be any of the BSON data types, including arrays and other documents.
3. **Database:** A database is a container for collections, which holds all the data for a particular application.
4. **Shard:** A shard is a horizontal partition of data that is spread across multiple servers. MongoDB uses sharding to distribute data across multiple nodes and support horizontal scalability.
5. **Replica Set:** A replica set is a group of servers that maintain identical copies of the same data. Replication provides high availability, durability, and data redundancy.
6. **Query Router:** Query Routers are the components of a MongoDB cluster that receive client requests, determine the location of the required data, and direct the request to the appropriate shard.

7. **Configuration Servers:** Configuration servers store metadata and configuration settings for a MongoDB cluster. They are responsible for maintaining the mapping of shards to replica sets.

In summary, MongoDB is widely used in modern web and mobile applications that need to handle large amounts of rapidly changing data in real time.

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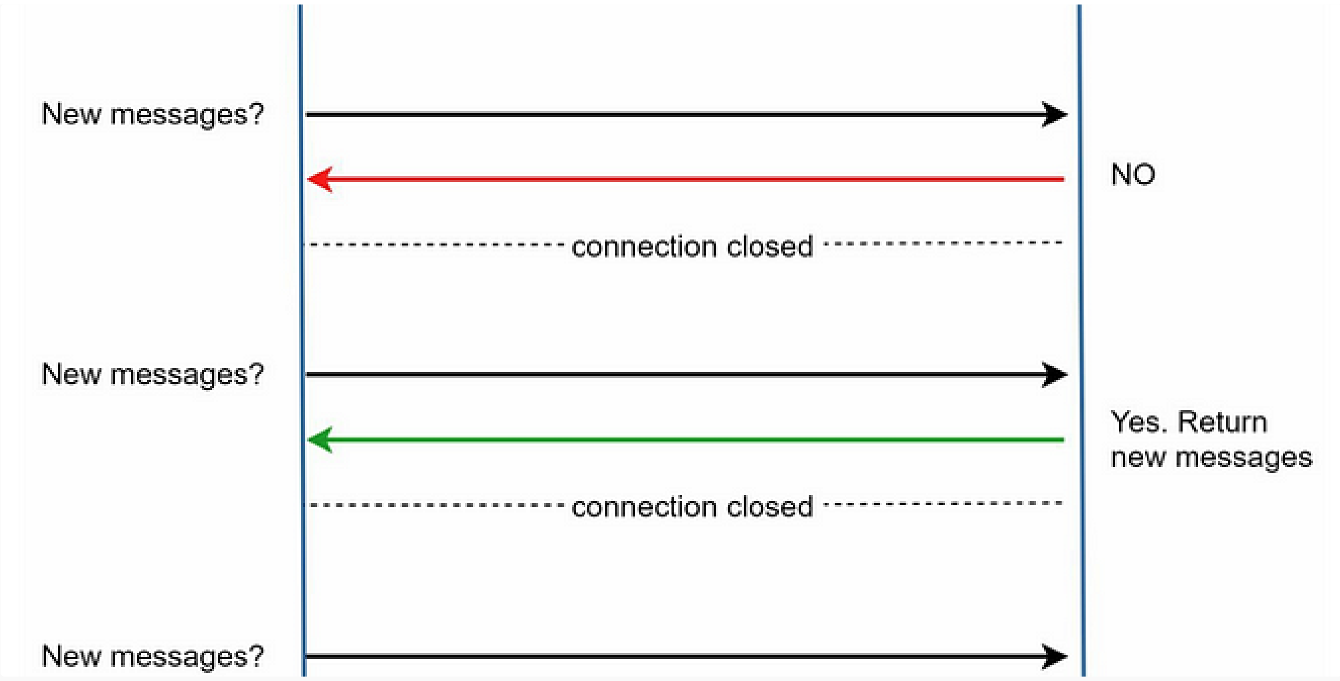

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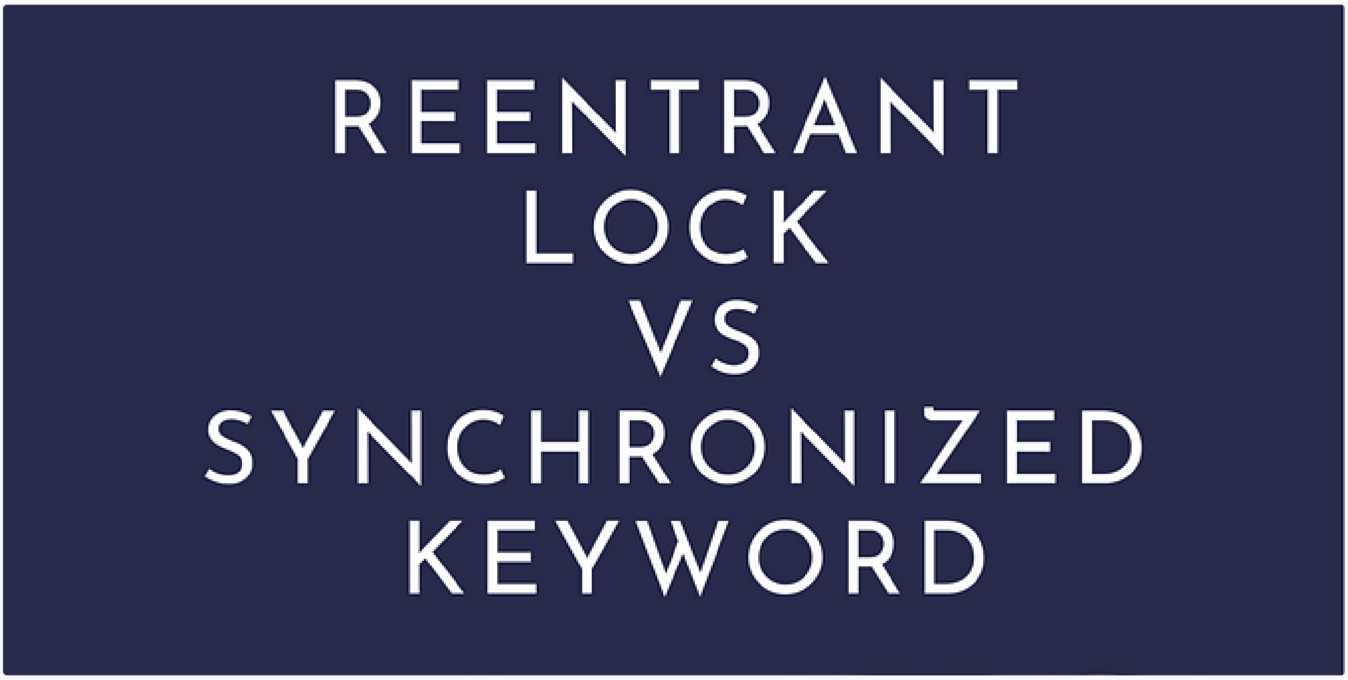

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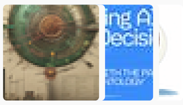

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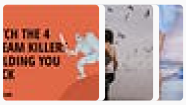

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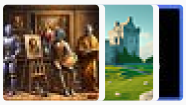

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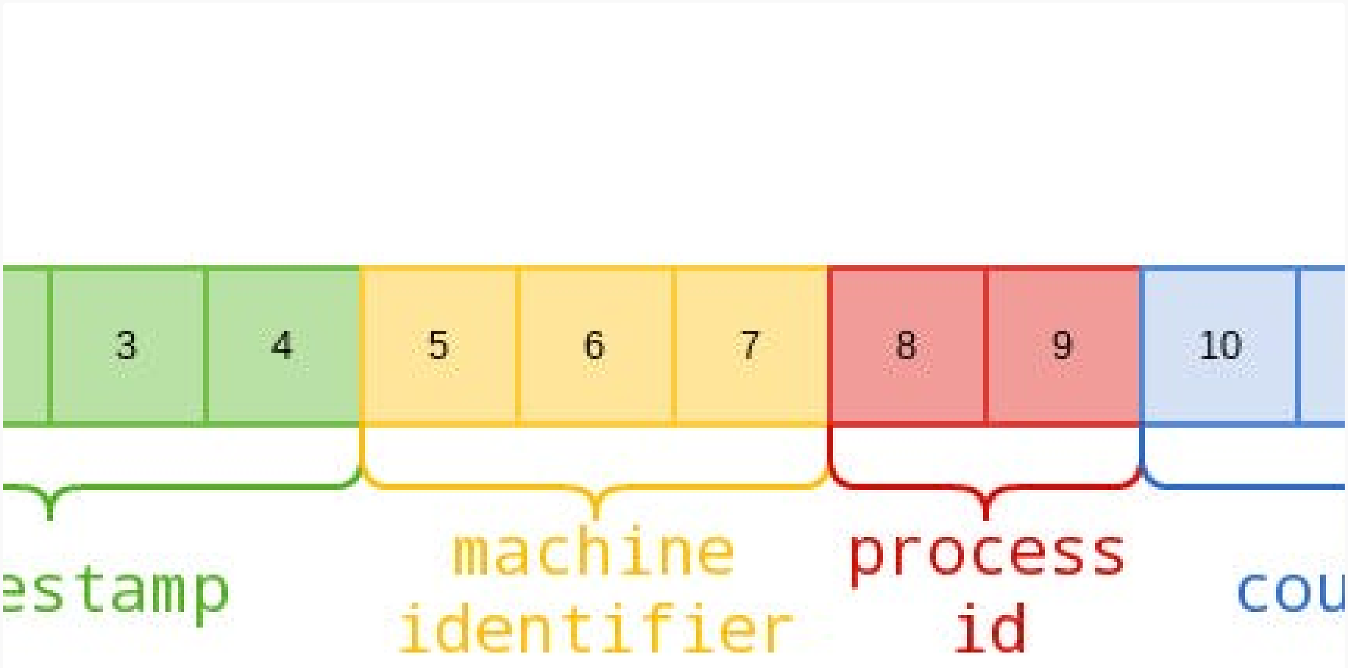

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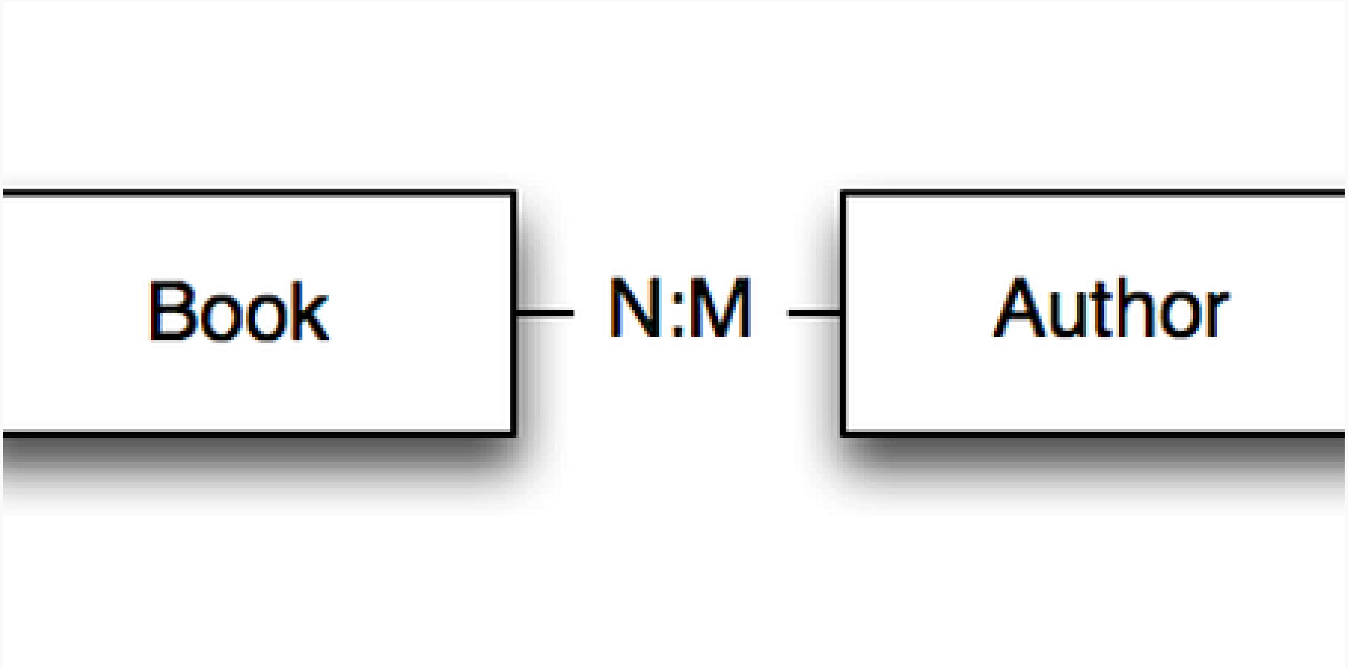

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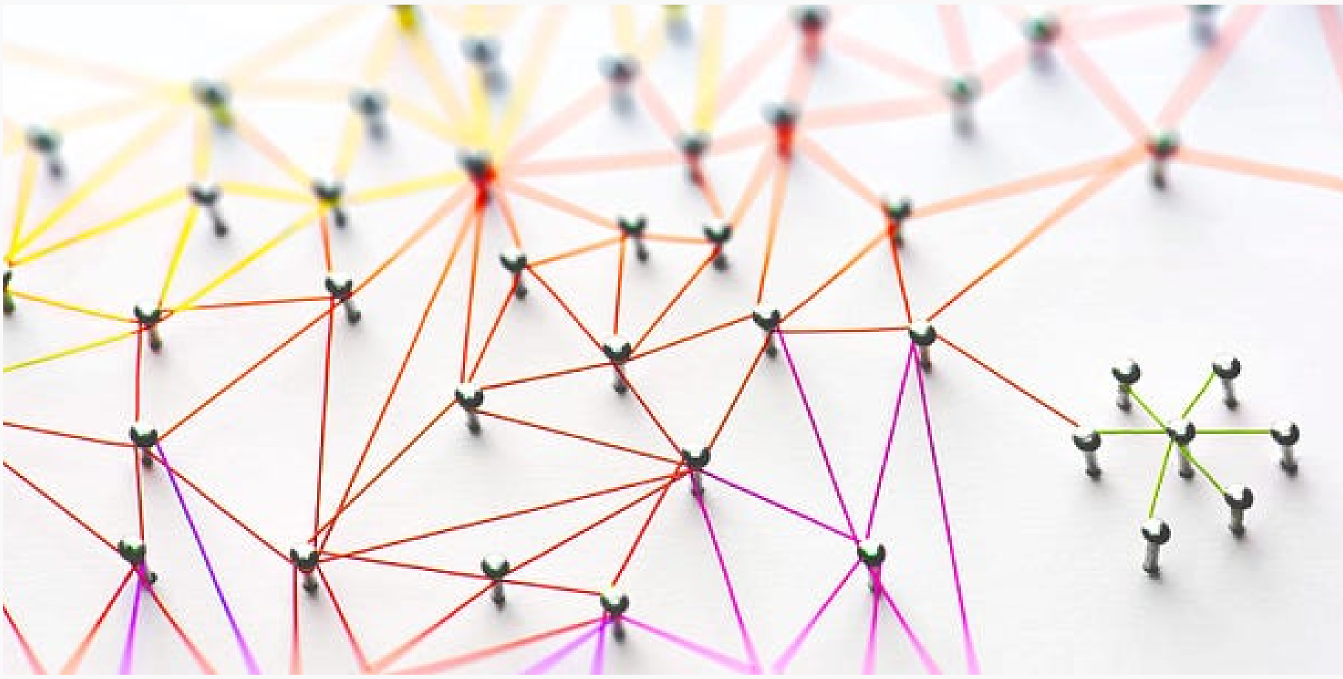

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